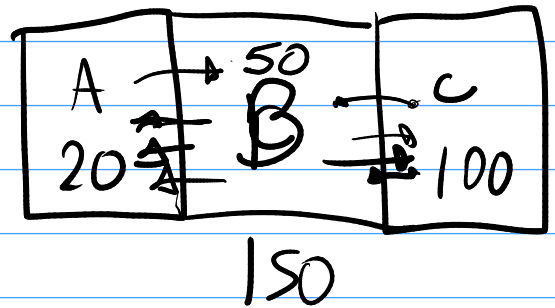
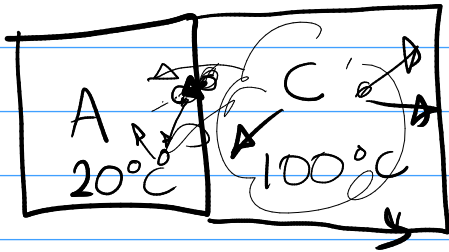
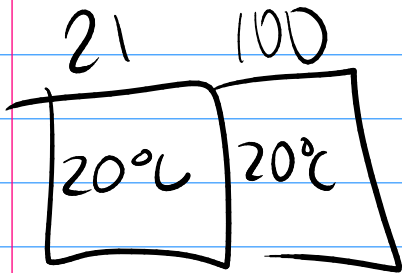


Calor



Ley cero de la Termodinámica



Calor (Q)

Capacidad calorífica

$$Q = C \Delta T$$

$\downarrow$
 $\frac{J}{^{\circ}C/K}$

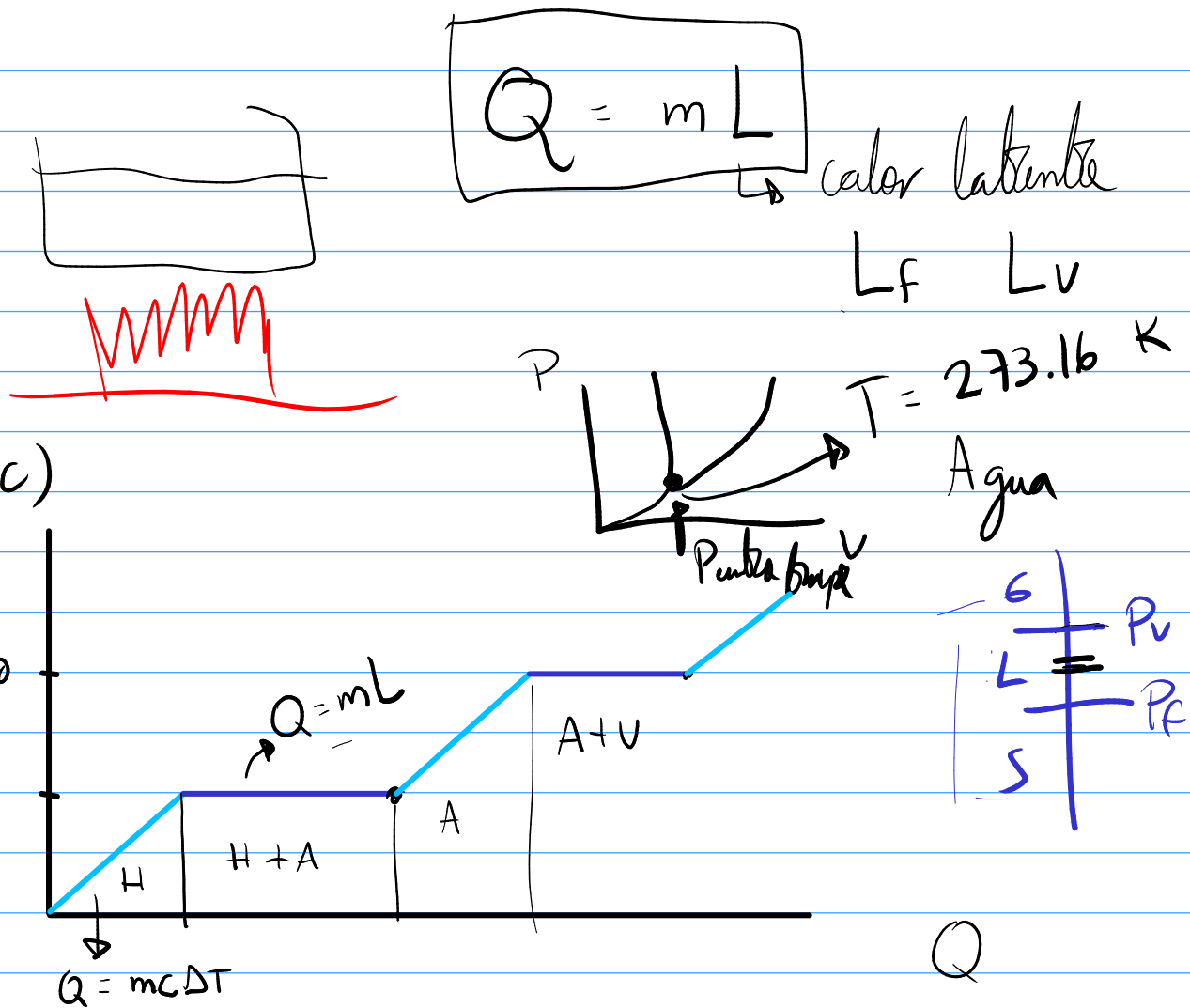
$$C = mc$$

$\downarrow$
 $\frac{J}{Kg^{\circ}C}$

$$Q = mc \Delta T$$

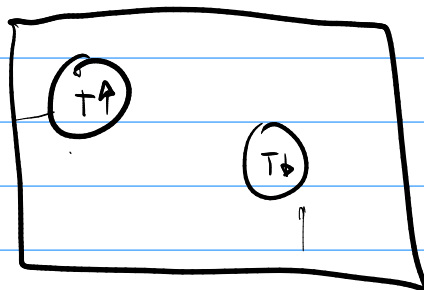
$\rightarrow$ calor de cambio de temperatura

$$\frac{100 J}{Kg^{\circ}C}$$



$$C_{H_2O} = 4186 \frac{J}{kg^{\circ}C}$$

Calorimetria



$$\Sigma Q_g + \Sigma Q_p = 0$$

Transferencia de calor

→ Conducción

→ Convección

→ Radiación

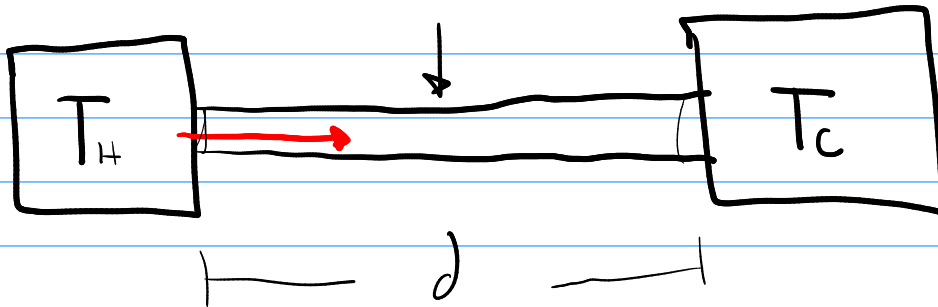
Conducción

$$\dot{Q} = \frac{Q}{t}$$

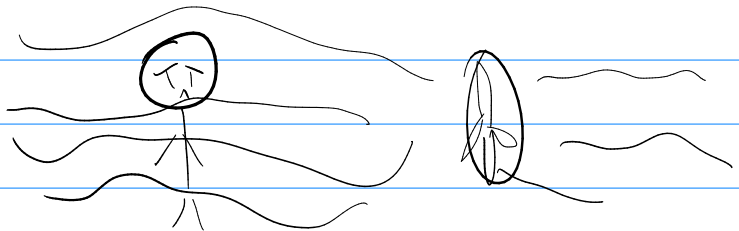
$$\dot{Q} = - \frac{kA \Delta T}{d}$$

$$\dot{Q} = \frac{[k][A](T_H - T_C)}{d}$$

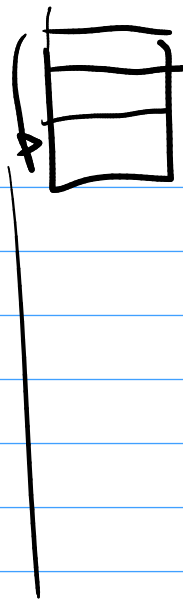
→ conductividad térmica



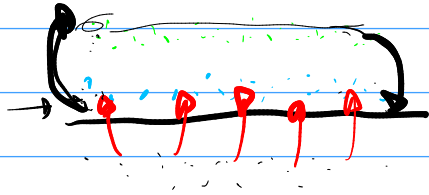
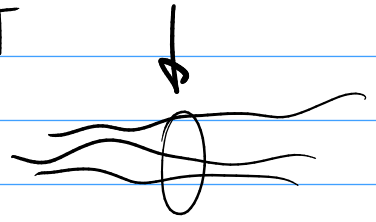
Convección



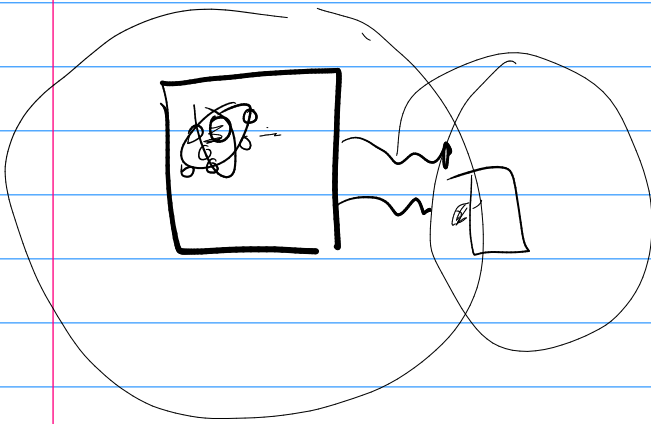
CN



CF



Radiación



$$\dot{Q} = \sigma \epsilon A (T^4 - T_0^4)$$

↙ ambiente

Constante de Stefan-Boltzmann

$$\sigma = 5.67 \times 10^{-8} \text{ J/s.m}^2\text{K}^4$$

$\epsilon \rightarrow$ emisividad

0

1 $\rightarrow$ "cuerpos negros"

A $\rightarrow$ área superficial

